Mohammad Sayeef Alam

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Summary

R Developer and Biostatistician with a PhD in Genetic Epidemiology and 6+ years of experience integrating complex biological data into actionable clinical insights. Proven expertise in building highly reproducible R packages, interactive R/Shiny applications, and automated bioinformatic pipelines (Snakemake). Experienced navigating strict regulatory frameworks (GxP) and utilizing CDISC standards (SDTM, ADaM) within end-to-end drug development workflows.

Experience

Graduate Research Assistant in HTA, PenTAG, University of Exeter – UK March 2026 – Present

- Evaluated industry-submitted Single Technology Appraisal (STA) models for NICE reimbursement by appraising economic literature, structural model, and sensitivity analyses.
- Engineered de novo HTA decision models and adapted existing frameworks using R and Excel VBA for Multiple Technology Appraisal (MTA) projects.
- Executed advanced statistical analyses spanning survival analysis, evidence generation, uncertainty quantification, and Delphi studies, delivering tailored solutions for external consultancy clients.

Researcher, NTNU – Trondheim, Norway April 2025 – March 2026

- Integrated and harmonized real-world data from Norway's patient registry, longitudianl health surveys, and biobanks into analysis-ready, linked datasets to support large-scale population health studies.
- Executed regression and longitudinal analyses on Norwegian registry and clinical data, applying advanced confounding control methods to derive robust causal evidence.
- Prototyped a modular R Shiny tool for Nordic HTA QALY-shortfall analysis, significantly reducing ad-hoc data requests from external partners.

PhD Candidate, NTNU – Trondheim, Norway April 2022 – March 2026

- Automated bioinformatic pipelines using Snakemake, eliminating repetitive processing across massive genomic datasets.
- Executed large-scale statistical pipelines (GWAS, polygenic risk scoring) across 24 million variants utilizing R, Python, and terminal environments.
- Integrated high-volume data from the HUNT biobank, health surveys, and clinical registries into analysis-ready formats.
- Authored peer-reviewed publications and led cross-functional technical collaborations with molecular biologists and clinical researchers.

Associate, Parexel – Remote, India July 2021 – February 2022

- Developed dynamic R Shiny applications for economic evaluations to support complex HTA submissions and strategic payer decision-making.
- Conducted thorough feasibility assessments, indirect treatment comparisons, and complete economic analyses for direct integration into HTA dossiers and regulatory/payer submissions.
- Designed and implemented interactive dashboards visualizing key clinical endpoints, including survival outcomes, treatment sequences, and demographic summaries.
- Delivered high-impact actionable insights to internal and external stakeholders, iteratively incorporating cross-functional feedback to refine final project deliverables under strict deadlines.

Education

PhD in Genetic Epidemiology April 2022 – March 2026
Norwegian University of Science and Technology – Trondheim, Norway

MPhil in Biostatistics and Demography July 2020 – May 2021
International Institute for Population Sciences – Mumbai, India

MSc in Biostatistics and Demography July 2018 – June 2020
International Institute for Population Sciences – Mumbai, India

BSc in Statistics April 2014 – May 2018
University of Calcutta – Kolkata, India

Skills

Languages & Tech:	R, Python, Bash/Terminal, SQL, SAS, Excel (VBA)
Software Engineering:	R/Shiny, R Package Development, Git Version Control, Docker, Snakemake (Nextflow equivalent), Reproducible Pipelines
Clinical & Regulatory:	CDISC (SDTM, ADaM, SEND), TLF Programming, GxP Compliance, Medical Data Review, Pharmacovigilance, Clinical Trial Workflows
Bioinformatics:	GWAS, Genomics, Bioconductor, Polygenic Risk Scoring, Multi-omics Integration
Biostatistics & RWE:	Real-World Data Analysis, Regression, Survival Analysis, Causal Inference, Health Economics / HTA

References

References available on request.